

Sysmon Investigation 3

Scenario.

Your team has informed you that the adversary has managed to set up persistence on your endpoints as they continue to move throughout your network. Find how the adversary managed to gain persistence using logs provided.

- Logs are located in “C:\Users\user\Documents\TryHackMe!\Sysmon”

Questions

What is the IP of the suspected adversary in Investigation 3.1?

For this investigation I used the event viewer tool built in windows.

I was looking for IP address I start looking for code 3 IDs logs for network connection

Información	2/12/2018 3:15:53 AM	Microsoft-Windows-Sysmon	3 (3)
Información	2/12/2018 3:15:58 AM	Microsoft-Windows-Sysmon	3 (3)
Información	2/12/2018 3:15:59 AM	Microsoft-Windows-Sysmon	3 (3)

Inside of the logs review the connections to find the IP address

Protocol tcp
Initiated true
SourceIsIpv6 false
SourceIp 172.16.199.179
SourceHostname DESKTOP-O153T4R.localdomain
SourcePort 54922
SourcePortName
DestinationIsIpv6 false
DestinationIp 172.30.1.253
DestinationHostname empirec2
DestinationPort 80

Reviewing the log I can extract several clues to claim that destination IP is malicious let's break down

- The source IP has the hostname “**DESKTOP-O153T4R**”, this name suggests a local machine or host
- The destination IP has the Destination Hostname “empirec2”, usually the c2 server are well known as malicious command and control

So based in these IOC I can claim thar the malicious IP address is “**172.30.1.253**”.

Question 2.

What is the hostname of the affected endpoint in Investigation 3.1?

Based on the question above where I already answer the question, the name host is “**DESKTOP-O153T4R**”

Question 3.

What is the hostname of the C2 server connecting to the endpoint in Investigation 3.1?

This one is pretty easy once we have the last 2 questions, with the review in the fist question the answer is “**empirec2**”.

Question 4.

Where in the registry was the payload stored in Investigation 3.1?

to answer this question first I reviewed the logs 13 ID because this code storage the changes made in registry key values

Id. del evento	Catego
3	(3)
12	(12)
12	(12)
12	(12)
13	(13)
13	(13)
18	(18)

Inside I check for any clue and found the following



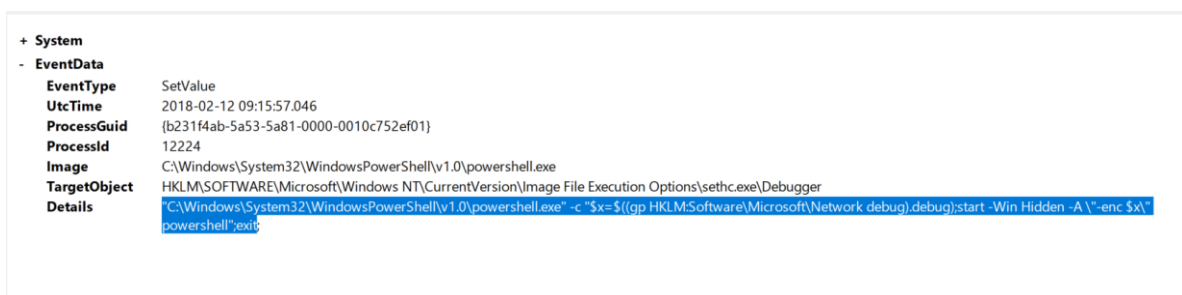
With this information I can build a strong hypothesis let's break down the log

- **Event type** set value: This means that this log is logging when new registry key was modified or altered with a new value stored in it
- **Image C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe:** This shows the process that made the modification in the registry key, we see power shell process that suggests a malicious activity
- **Target object** HKLM\SOFTWARE\Microsoft\Network\debug: This is the path where the change was done
- **Target object** HKLM\SOFTWARE\Microsoft\Network\debug: This are exactly what we are looking for, the path where the registry key was modified for storage the payload
- **Details:** This is payload that was storage in the registry key probably a power shell script encoded in base 64 by the threat actor.

Question 4.

What PowerShell launch code was used to launch the payload in Investigation 3.1?

I checked another one log id 13 and found this one.



Here I can see several IOC let's analyst the most important

- **Image:** the process used to launch the script
- **Target object:** the path where the registry key will be modified by the script
- **Details:** the payload used by the attacker and our answer